# What is Counting?

Counting is a part of mathematics that helps us find **how many ways** something can happen. Instead of listing every possibility, we use some smart rules to count quickly.
**Think of it like this:** If you want to know how many different outfits you can make, you don't need to try on every combination. You can use math to find out!

## The Two Main Rules

There are two important rules that make counting easy:

- Sum Rule
- Product Rule

# Sum Rule

## What does it mean?

If an action has **multiple options** and you do **only one**, then add the number of choices.

## Key Idea:

- Use when you have "OR" situations
- You can pick from Group A **OR** Group B **OR** Group C
- You cannot pick from multiple groups at the same time
- You're selecting from different categories
- Only one selection is made
- Keywords: "or", "either", "choose one from"

# Product Rule

## What does it mean?

If an action has **multiple options**, and you have to do **all** of them, then multiply the number of choices for each step.

## Key Idea:

- Use when you have "AND" situations
- You need to pick from Group A **AND** Group B **AND** Group C
- Each choice is independent of others
- You're making multiple selections
- Each step requires a choice

- Keywords: "and", "then", "followed by"

# Practice Problem

### Problem 1:

You have 3 red pens and 4 blue pens. How many ways can you pick one pen?

### Solve:

Red pens = 3
Blue pens = 4
Total ways = Red pens + Blue pens
Total ways = 3 + 4 = 7

### Detailed Explanation:

- You want to pick ONE pen
- You can pick a red pen OR a blue pen
- Red pens: 3 choices
- Blue pens: 4 choices
- Since it's "OR", use Sum Rule: 3 + 4 = 7

### Problem 2:

You flip a coin and then roll a dice. How many different outcomes are possible?

### Solve:

Coin outcomes = 2 (Heads, Tails)
Dice outcomes = 6 (1, 2, 3, 4, 5, 6)
Total outcomes = Coin outcomes × Dice outcomes
Total outcomes = 2 × 6 = 12

### Detailed Explanation:

- First step: Flip coin (2 outcomes: Heads OR Tails)
- Second step: Roll dice (6 outcomes: 1, 2, 3, 4, 5, 6)
- You do BOTH steps, so use Product Rule
- Total outcomes = 2 × 6 = 12
- Examples: (Heads,1), (Heads,2), ..., (Tails,6)

## Problem 3:

A pizza place offers small or medium or large pizzas. Small comes in 2 flavors, medium in 3 flavors, and large in 5 flavors. How many different pizzas can you order?

## Solve:

Small pizzas = 2 flavors
Medium pizzas = 3 flavors
Large pizzas = 5 flavors
Total pizzas = Small + Medium + Large
Total pizzas = 2 + 3 + 5 = 10

## Detailed Explanation:

- You choose ONE pizza size, then ONE flavor for that size
- Small pizzas: 2 options
- Medium pizzas: 3 options
- Large pizzas: 5 options
- Since you pick one category, use Sum Rule: 2 + 3 + 5 = 10

## Problem 4:

A class has 5 boys and 3 girls. How many ways can you choose a 2-person committee with exactly 1 boy and 1 girl?

## Solve:

Ways to choose 1 boy from 5 boys = 5
Ways to choose 1 girl from 3 girls = 3
Total ways = Boys × Girls
Total ways = 5 × 3 = 15

## Detailed Explanation:

- You need to pick 1 boy AND 1 girl
- Ways to pick 1 boy from 5 boys: 5 ways
- Ways to pick 1 girl from 3 girls: 3 ways
- Since you need BOTH selections, use Product Rule: 5 × 3 = 15

## Problem 5:

To go from point A to point C, you must pass through point B. There are 4 roads from A to B and 3 roads from B to C. How many different routes are there from A to C?

## Solve:

Roads from A to B = 4
Roads from B to C = 3
Total routes = (A to B) × (B to C)
Total routes = 4 × 3 = 12

## Detailed Explanation:

- This is a two-stage journey: A→B, then B→C
- First stage: A to B (4 choices)
- Second stage: B to C (3 choices)
- You must complete BOTH/ALL stages, so use Product Rule
- Total routes = 4 × 3 = 12

## Problem 6:

To get to work, you can either:

- Take the subway (3 different lines), then walk (2 different paths)
- Take a bus (4 different routes) directly
- Drive your car (1 way) or borrow a friend's car (2 friends available)

How many different ways can you get to work?

## Solve:

Method 1: Subway + Walk
Subway lines = 3, Walk paths = 2
Subway + Walk ways = 3 × 2 = 6 (Product Rule)

Method 2: Bus directly
Bus ways = 4

Method 3: Drive
Your car = 1, Friend's car = 2
Drive ways = 1 + 2 = 3 (Sum Rule)

Total ways = Method 1 + Method 2 + Method 3 (Sum Rule)
Total ways = 6 + 4 + 3 = 13

## Detailed Explanation:

Method 1: Subway + Walk

- Take subway (3 lines) AND then walk (2 paths)
- Use Product Rule: $3 \times 2 = 6$ ways

Method 2: Bus directly

- Take bus: 4 ways

Method 3: Drive

- Your car OR friend's car
- Use Sum Rule: $1 + 2 = 3$ ways

Total ways:

- You choose ONE way
- All methods are alternatives (OR), so use Sum Rule, Total = $6 + 4 + 3 = 13$ ways

## Problem 7:

How many 4-letter words can be formed using the letters A, B, C, D, E if:

- The first letter must be A or B
- The last letter cannot be E
- Letters can be repeated

### Solve:

Position 1 (First): A or B = 2 choices
Position 2 (Second): A, B, C, D, E = 5 choices
Position 3 (Third): A, B, C, D, E = 5 choices
Position 4 (Last): A, B, C, D (not E) = 4 choices

Total words = Position 1 × Position 2 × Position 3 × Position 4
Total words = $2 \times 5 \times 5 \times 4 = 200$

### Detailed Explanation:

- Position 1 (First letter): A OR B = $1 + 1 = 2$ choices
- Position 2 (Second letter): Any of the 5 letters (A OR B OR C OR D OR E) $=1+1+1+1+1 = 5$ choices
- Position 3 (Third letter): Any of the 5 letters (A OR B OR C OR D OR E) $=1+1+1+1+1 = 5$ choices

- Position 4 (Last letter): Any letter except E (A OR B OR C OR D )= 1+1+1+1= 4 choices
- Since we need **ALL** positions filled, use Product Rule: Total = 2 × 5 × 5 × 4 = 200 words

**Problem 8:**

A student can take either Math 101 or Math 102 in the morning, and either Physics 201, Chemistry 301, or Biology 401 in the afternoon. How many different schedules are possible?

### Solve:

Morning classes = 2 choices (Math 101 or Math 102)
Afternoon classes = 3 choices (Physics, Chemistry, or Biology)
Total schedules = Morning × Afternoon Total schedules = 2 × 3 = 6

### Detailed Explanation:

- Morning: Choose 1 math class from 2 options
- Afternoon: Choose 1 science class from 3 options
- You need BOTH morning AND afternoon classes, so use Product Rule
- Total schedules = 2 × 3 = 6
- Possible schedules: (Math 101, Physics), (Math 101, Chemistry), (Math 101, Biology), (Math 102, Physics), (Math 102, Chemistry), (Math 102, Biology)

**Problem 9:**

A security system requires a 3-digit code where each digit can be 0-9, but the first digit cannot be 0. How many possible codes are there?

### Solve:

First digit = 9 choices (1, 2, 3, 4, 5, 6, 7, 8, 9)
Second digit = 10 choices (0, 1, 2, 3, 4, 5, 6, 7, 8, 9)
Third digit = 10 choices (0, 1, 2, 3, 4, 5, 6, 7, 8, 9)
Total codes = First × Second × Third Total codes = 9 × 10 × 10 = 900

### Detailed Explanation:

- First digit: Cannot be 0, so we have 1, 2, 3, 4, 5, 6, 7, 8, 9 = 9 choices
- Second digit: Can be any digit 0-9 = 10 choices
- Third digit: Can be any digit 0-9 = 10 choices
- Since we need ALL three positions filled, use Product Rule: 9 × 10 × 10 = 900 codes

**Problem 10:**

How many license plates can be made using 2 letters followed by 3 digits, where the first letter must be A, B, or C, and the last digit must be even?

## Solve:

Position 1 (First letter): A, B, or C = 3 choices
Position 2 (Second letter): A-Z = 26 choices
Position 3 (First digit): 0-9 = 10 choices
Position 4 (Second digit): 0-9 = 10 choices
Position 5 (Last digit): 0, 2, 4, 6, 8 = 5 choices (even digits)
Total license plates = 3 × 26 × 10 × 10 × 5 = 39,000

## Detailed Explanation:

- Position 1: Must be A, B, or C = 3 choices
- Position 2: Can be any letter A-Z = 26 choices
- Position 3: Can be any digit 0-9 = 10 choices
- Position 4: Can be any digit 0-9 = 10 choices
- Position 5: Must be even (0, 2, 4, 6, 8) = 5 choices
- Since ALL positions must be filled, use Product Rule: 3 × 26 × 10 × 10 × 5 = 39,000 plates

**Problem 11:**

A survey asks people to choose their favorite sport from football, basketball, or tennis, and their favorite season from spring, summer, fall, or winter. However, if someone chooses tennis, they cannot choose winter. How many different response combinations are possible?

## Solve:

Case 1:
Choose football or basketball Sports = 2 choices (football or basketball)
Seasons = 4 choices (any season)
Case 1 combinations = 2 × 4 = 8

Case 2:
Choose tennis Sports = 1 choice (tennis)
Seasons = 3 choices (spring, summer, fall - not winter)
Case 2 combinations = 1 × 3 = 3

Total combinations = Case 1 + Case 2
Total combinations = 8 + 3 = 11

## Detailed Explanation:

- Case 1: Football or Basketball
    - Choose sport (football OR basketball) = 2 choices
    - Choose season (any of the 4) = 4 choices
    - Use Product Rule: 2 × 4 = 8 combinations
- Case 2: Tennis

- Choose sport (tennis only) = 1 choice
- Choose season (spring, summer, fall - winter not allowed) = 3 choices
- Use Product Rule: $1 \times 3 = 3$ combinations

- These cases are mutually exclusive (OR situations) , Use Sum Rule: $8 + 3 = 11$ total response combinations

**Problem 12:**

A phone number has the format XXX-XXXX where X represents a digit. The first digit cannot be 0 or 1, and the fourth digit (first digit of the last four) cannot be 0. How many possible phone numbers are there?

## Solve:

Position 1: 2, 3, 4, 5, 6, 7, 8, 9 = 8 choices
Position 2: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 = 10 choices
Position 3: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 = 10 choices
Position 4: 1, 2, 3, 4, 5, 6, 7, 8, 9 = 9 choices
Position 5: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 = 10 choices
Position 6: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 = 10 choices
Position 7: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 = 10 choices
Total phone numbers = $8 \times 10 \times 10 \times 9 \times 10 \times 10 \times 10 = 7{,}200{,}000$

## Detailed Explanation:

- Since ALL positions must be filled, use Product Rule: $8 \times 10 \times 10 \times 9 \times 10 \times 10 \times 10 = 7{,}200{,}000$

**Problem 13:**

A computer password must be exactly 6 characters long and can contain uppercase letters, lowercase letters, or digits. However, it must contain at least one digit. How many passwords are possible?

## Solve:

Total possible characters = $26 + 26 + 10 = 62$ (uppercase + lowercase + digits)

Step 1:
Total passwords with any characters Each position has 62 choices
Total with any characters = $62^6 = 56{,}800{,}235{,}584$

Step 2:
Passwords with NO digits (only letters) Total letters = $26 + 26 = 52$
Passwords with only letters = $52^6 = 19{,}770{,}609{,}664$

Passwords with at least one digit = Total - Passwords with no digits
Passwords with at least one digit = $56{,}800{,}235{,}584 - 19{,}770{,}609{,}664 = 37{,}029{,}625{,}920$

## Detailed Explanation:

- We use complement counting: find total possibilities, then subtract unwanted cases
- Total characters available: 26 uppercase + 26 lowercase + 10 digits = 62 characters
- Each of the 6 positions can be any of these 62 characters
- Total passwords (any characters): $62 \times 62 \times 62 \times 62 \times 62 \times 62 = 62^6$
- Unwanted case: passwords with NO digits (only letters)
- Letters available: 26 uppercase + 26 lowercase = 52 letters
- Passwords with only letters: $52^6$
- Answer = Total passwords - Passwords with no digits
- This gives us passwords that must have at least one digit

**Problem 14:**

A complete binary tree has height 3 (levels 0, 1, 2, 3). You want to create a path from the root to any leaf node. At each internal node, you can go left or right. However, if you go left at level 1, you cannot go right at level 2. How many different valid paths from root to leaf are possible?

## Solve:

Case 1: Go RIGHT at level 1
Level 1: Right = 1 choice
Level 2: Left or Right = 2 choices
Level 3: Left or Right = 2 choices
Case 1 paths = $1 \times 2 \times 2 = 4$

Case 2: Go LEFT at level 1, then LEFT at level 2
Level 1: Left = 1 choice
Level 2: Left only = 1 choice (restriction: cannot go right)
Level 3: Left or Right = 2 choices
Case 2 paths = $1 \times 1 \times 2 = 2$

Total valid paths = Case 1 + Case 2 = 4 + 2 = 6

## Detailed Explanation:

- In a complete binary tree of height 3, we need to make 3 decisions to reach a leaf
- Normal case: each level offers 2 choices (left OR right)
- But we have a restriction: if left at level 1, then cannot go right at level 2
- Case 1: Right at level 1
  - Level 1: Choose right = 1 way
  - Level 2: Choose left OR right = 2 ways (no restriction applies)
  - Level 3: Choose left OR right = 2 ways
  - Use Product Rule: $1 \times 2 \times 2 = 4$ paths
- Case 2: Left at level 1

    - Level 1: Choose left = 1 way
    - Level 2: Must choose left (restriction) = 1 way
    - Level 3: Choose left OR right = 2 ways
    - Use Product Rule: $1 \times 1 \times 2 = 2$ paths
- These cases are mutually exclusive (OR), so use Sum Rule: $4 + 2 = 6$ total valid paths